Some definitions

- Sampling frequency (of an A/D): Inverse of the sampling period $F_s = 1/T_s$, while T is the interval between samples.
- Folding frequency: $F_s/2$
- Nyquist rate: minimum sampling rate of an analog signal to avoid aliasing $F_s > 2F_{max}$.
- Quantization step (Δ): $\frac{x_{max} - x_{min}}{L-1}$, where L is the number of quantization levels

Sampling and Quantization

A digital communication link carries binary-coded words representing samples of an input signal:

$$x_a(t) = 3\cos(600\pi t) + 2\cos(1800\pi t)$$

The link is operated at 10,000 bits/s and each input sample is quantized into 1024 different voltage levels.

- What are the sampling frequency and folding frequency?

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5. What is the Nyquist rate for the signal $x_a(t)$

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- What are the frequencies in the resulting discrete-time signal $x(n)$

Sampling and Quantization

A digital communication link carries binary-coded words representing samples of an input signal:

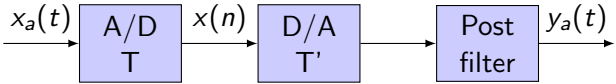
$$x_a(t) = 3\cos(600\pi t) + 2\cos(1800\pi t)$$

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d. What is the resolution Δ?

Seat Work (can be done by pairs)

Consider the simple signal processing system as shown below. The sampling periods of the A/D and the D/A are $T = 5$ ms and $T' = 1$ ms, respectively. The post-filter removes any frequency component above $F_s'/2$.



- Determine the output ($y_a(t)$) of the system if the input is:

$$x_a(t) = 3\cos(100\pi t) + 2\sin(250\pi t)$$

- The discrete-time signal $X(n) = 6.5\cos(\pi n/10)$ is quantized with a resolution (a) $\Delta = 0.1$ or (b) $\Delta = 0.02$. How many bits are required in the A/D converter in each case?